

Goldfish Architecture: Therapeutic Urban Aquaponics

Alex Orsholits
alex-orsholits@g.ecc.u-tokyo.ac.jp
The University of Tokyo
Japan

Zhou Li
lizhouzhouli@g.ecc.u-tokyo.ac.jp
The University of Tokyo
Japan

Kritika Dhupar
kritikadhupar@g.ecc.u-tokyo.ac.jp
The University of Tokyo
Japan

Yusuke Obuchi
obuchi@arch1.t.u-tokyo.ac.jp
The University of Tokyo
Japan

Chengzhe Zhu
chengzhe@g.ecc.u-tokyo.ac.jp
The University of Tokyo
Japan

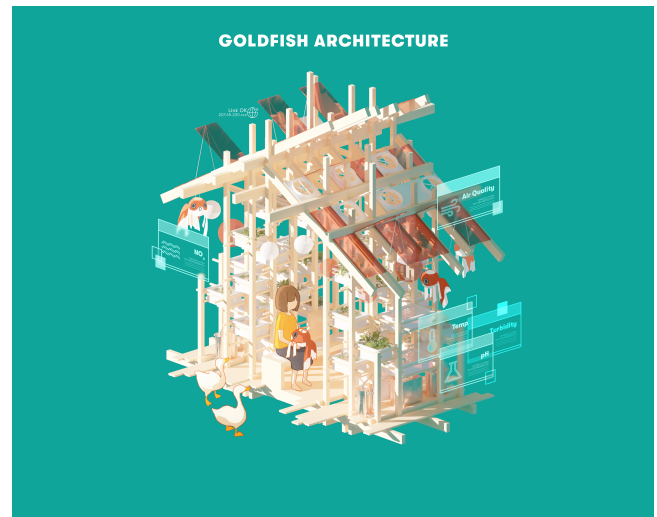
Yusuke Komata
komata@arch1.t.u-tokyo.ac.jp
The University of Tokyo
Japan

Shuta Takagi
mail@takagishuta.com
The University of Tokyo
Japan

ABSTRACT

In recent years, there has been a transition towards new implementations of digital augmentation and optimization at everincreasing scales. With metropolises now being equipped with a myriad of sensors, these agglomerations are becoming entangled with a new class of infrastructure. Yet, current propositions of big data-based sustainability implementations leave much to be desired in terms of transparency. To the detriment of personal and interpersonal comfort, these configurations operate solely as city-wide optimizers, on a scale that is alienating for the individual. The benefits therefore remain intangible in the eyes of the inhabitants. Could these benefits operate on a smaller, personal scale to a similar effect as that of a larger scale? In a fundamental sense, current environment & sustainability implementations represent a disconnect from the individual needs of the community as the process falls solely on the system developer, rather than the public willing to accommodate such changes in their space.

For the 2021 Seoul Biennale of Architecture and Urbanism: Crossroads - Building the resilient city, the University of Tokyo Department of Architecture Advanced Studies Program (T_ADS) sought to promote appreciation and compassion for our personal environments through the collective care of plants and fish, all while providing people with the therapeutic benefits from the process of caring for pets and the satisfaction of harvesting ones own produce. While the goal of the laboratory's production in the case of the Biennale is the design and construction of an architectural prototype encapsulating two remotely manageable aquaponic systems that seek to provide urban communities a symbiotic relationship between fish, plants, and users, for SIGGRAPH Asia, we intend to take advantage of the unique architectural expression of virtual environments, which we treat with equal importance to that of



physical reality, to design a Digital Twin closely linked with the constructed prototype.

ACM Reference Format:

Alex Orsholits, Zhou Li, Kritika Dhupar, Yusuke Obuchi, Chengzhe Zhu, Yusuke Komata, and Shuta Takagi. 2021. Goldfish Architecture: Therapeutic Urban Aquaponics. In *SIGGRAPH Asia 2021 Art Gallery (SA '21 Art Gallery)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3476123.3487873>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '21 Art Gallery, December 14–17, 2021, Tokyo, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8686-9/21/12.

<https://doi.org/10.1145/3476123.3487873>